

typcas v2 CAS Suite

Test Case 1

Nested exponential + trig

$$e^{\sin(x^2)}$$

$$d/dx: x \cdot \cos(x^2) \cdot e^{\sin(x^2)} \cdot 2$$

$$f: C + \int e^{\sin(x^2)} dx$$

Test Case 2

Product of three terms

$$(x^2 + 1) \cdot (x - 3) \cdot (x + 2)$$

$$d/dx: (x^2 + 2(x \cdot (x - 3)) + 1) \cdot (x + 2) + (x^2 + 1) \cdot (x - 3)$$

$$f: C + \int (x^2 + 1) \cdot (x - 3) \cdot (x + 2) dx$$

Test Case 3

Quotient + log chain

$$\ln\left(\frac{x^2 + 1}{x - 1}\right)$$

$$d/dx: \frac{2(x \cdot (x - 1)) - 1 - x^2}{(x - 1)^2} \cdot \left(\frac{1}{\frac{x^2 + 1}{x - 1}}\right)$$

$$f: C + \int \ln\left(\frac{x^2 + 1}{x - 1}\right) dx$$

Test Case 4

Deeply nested powers

$$\left((x^2 + 1)^3 + x\right)^2$$

$$d/dx: 2\left(6\left(x \cdot (x^2 + 1)^2\right) + 1\right) \cdot \left(x + (x^2 + 1)^3\right)$$

$$f: \mathbb{C} + \int \left(x + (x^2 + 1)^3 \right)^2 dx$$

Test Case 5

Rational function

$$\frac{x^3 - 1}{x - 1}$$

$$d/dx: \frac{3(x^2 \cdot (x - 1)) + 1 - x^3}{(x - 1)^2}$$

$$f: \mathbb{C} + \int \frac{x^3 - 1}{x - 1} dx$$

Step Preview

$$\int 2x \cdot \cos(x^2) dx$$

$$\text{set } u_1 = x^2$$

$$\text{derive } du_1 = x \cdot 2$$

$$= \int 2x \cdot \cos(x^2) dx \Rightarrow \int \cos(u_1) du_1 \quad (\text{Apply u-substitution})$$

$$= \int \cos(u_1) du_1 \mapsto \sin(u_1) + C \quad (\text{Integrate in } u_1)$$

$$= \sin(u_1) + C \Rightarrow \sin(x^2) + C \quad (\text{Back-substitute})$$